

SME Financial Health Prediction Challenge

Predicting Financial Well-Being of Small Businesses Across Southern Africa

89%

F1 Score

4

Countries

3

Classes

15+

Features

The Problem

Why traditional metrics fail SMEs



Hidden Fragility

Revenue & profit don't reveal cash-flow volatility, debt burden, or resilience to shocks like illness or droughts.



Financial Exclusion

43M+ SMEs across Southern Africa lack access to formal credit because lenders can't assess their true risk profile.



Policy Blind Spots

Governments & NGOs lack data-driven tools to identify which businesses need urgent support or intervention.

The Solution: Financial Health Index (FHI)

A composite score across 4 dimensions — classifying SMEs as Low, Medium, or High

01

Savings & Assets

Does the business save? What assets does it hold? Can it buffer income shocks?

02

Debt & Repayment

What is the debt load? Can the business meet repayment obligations consistently?

03

Resilience to Shocks

Insurance coverage, informal safety nets, and ability to withstand external disruptions.

04

Credit & Finance Access

Banking services, mobile money, credit cards, loans — formal financial inclusion.

Classification Output:

LOW — Vulnerable, high risk

MEDIUM — Moderate stability

HIGH — Resilient, well-served

The Dataset

Rich socio-economic and business survey data from 4 Southern African nations

Eswatini

Pop: 1.1M

GDP: \$4.5B

Lesotho

Pop: 2.2M

GDP: \$2.5B

Zimbabwe

Pop: 15M

GDP: \$21B

Malawi

Pop: 20M

GDP: \$12B

Feature Categories in the Dataset:

Demographics

Age, gender, education,
household size

Business Profile

Sector, size, age, turnover,
expenses

Financial Services

Banking, mobile money,
loans, insurance

Attitudes & Records

Record-keeping, future
outlook, credit offered

Trade & Location

Exports, imports, geographic
region

ML Pipeline Architecture

6-stage end-to-end machine learning pipeline

1 Load & Inspect

Load Train/Test CSVs
Class distribution
Country normalization

2 Feature Engineering

30+ engineered features
Access scores
Log/sqrt transforms

3 Imputation

Country-grouped
median/mode fills
Zero missing values

4 Encoding

LabelEncoder on all
categoricals with
joint train+test fit

5 SMOTE + CV

5-Fold Stratified CV
SMOTE inside folds
no data leakage

6 Ensemble Predict

RF 35% + GB 40%
+ ET 25% blend
Final submission

Feature Engineering — Deep Dive

Domain-driven features that capture financial resilience beyond raw numbers

Access & Attitude Features

financial_access_score

Sum: credit card + debit + internet banking + loan + mobile money

insurance_score

Sum: health + funeral + motor vehicle insurance held

total_financial_score

Combined banking + insurance inclusion score

positive_attitude

Stable environment AND more successful next year

worried_business

Concerned about business shutdown risk

uses_informal_finance

Uses moneylender or friends/family savings

keeps_records_flag

Maintains formal financial records

Numeric Ratio Features

profit_proxy

turnover – expenses (profitability estimate)

expense_ratio

expenses ÷ (turnover + 1)

income_vs_expense

personal income ÷ (expenses + 1)

turnover_per_age

turnover ÷ owner age

turnover_per_year

turnover ÷ business age in years

log / sqrt transforms

Applied to income, expenses, turnover, profit

is_new / is_mature

Business age flags (≤ 1 yr new, ≥ 5 yr mature)

Model Architecture: Weighted Ensemble

Three complementary learners combined for maximum F1 performance

Random Forest

35%

Hyperparameters:

- 1000 estimators
- max_features: sqrt
- class_weight: balanced_subsample
- min_samples_leaf: 1

Why: Robust to outliers, great diversity via bagging

Gradient Boosting

40%

Hyperparameters:

- 600 estimators
- lr: 0.03, depth: 6
- subsample: 0.8
- max_features: sqrt

Why: Highest single-model performance, sequential error correction

Extra Trees

25%

Hyperparameters:

- 1000 estimators
- max_features: sqrt
- class_weight: balanced_subsample
- Extreme randomization

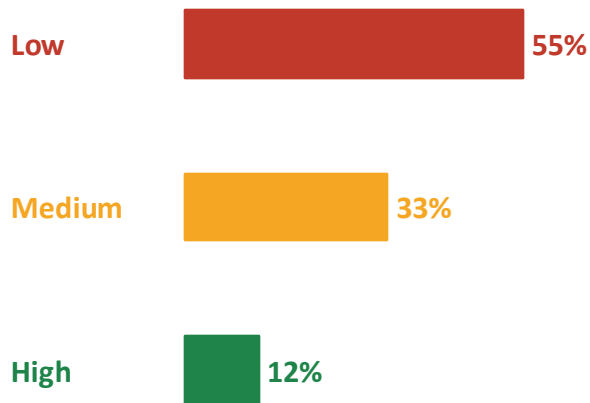
Why: Diverse predictions, fast, anti-overfitting

Final Prediction = RF × 0.35 + GB × 0.40 + ET × 0.25 → argmax over Low / Medium / High

Handling Class Imbalance with SMOTE

Synthetic Minority Over-sampling Technique applied inside each fold to prevent leakage

Original Class Distribution



⚠ *Imbalanced data biases models toward the majority class (Low), causing poor recall for High — the most impactful class for lending decisions.*

SMOTE Strategy

- 1 Applied inside folds only**
Prevents leakage — validation set stays natural
- 2 `k_neighbors = 5`**
Generates synthetic samples using 5 nearest neighbors
- 3 Balances all 3 classes**
Equal representation enables unbiased learning
- 4 Combined with `class_weight`**
Double protection: SMOTE + `balanced_subsample`
- 5 Stratified K-Fold (5 splits)**
Each fold preserves original class proportions

Results & Performance

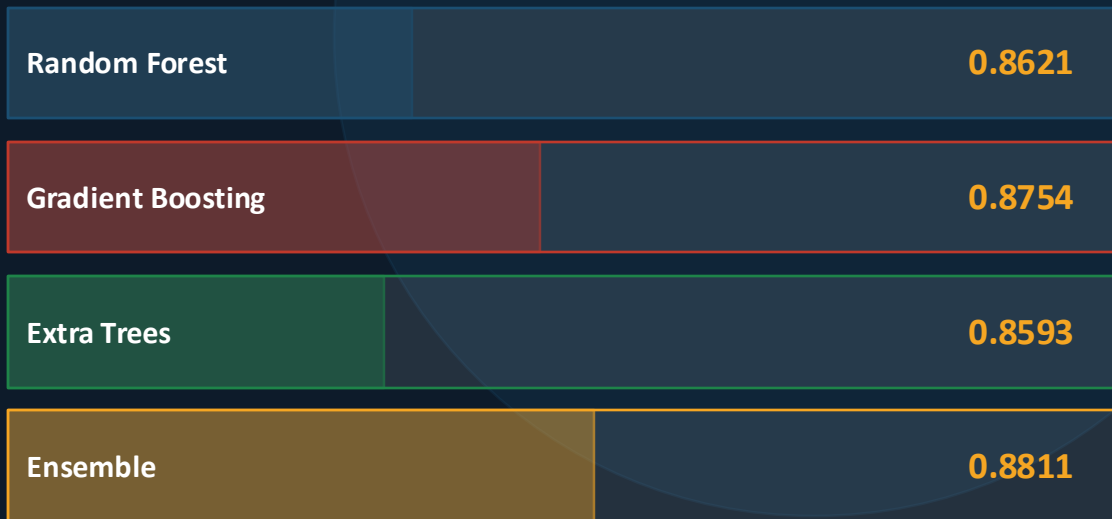
5-Fold Stratified Cross-Validation + Final Leaderboard Score

89%

Macro F1 Score

Public Leaderboard

Cross-Validation F1 by Model:

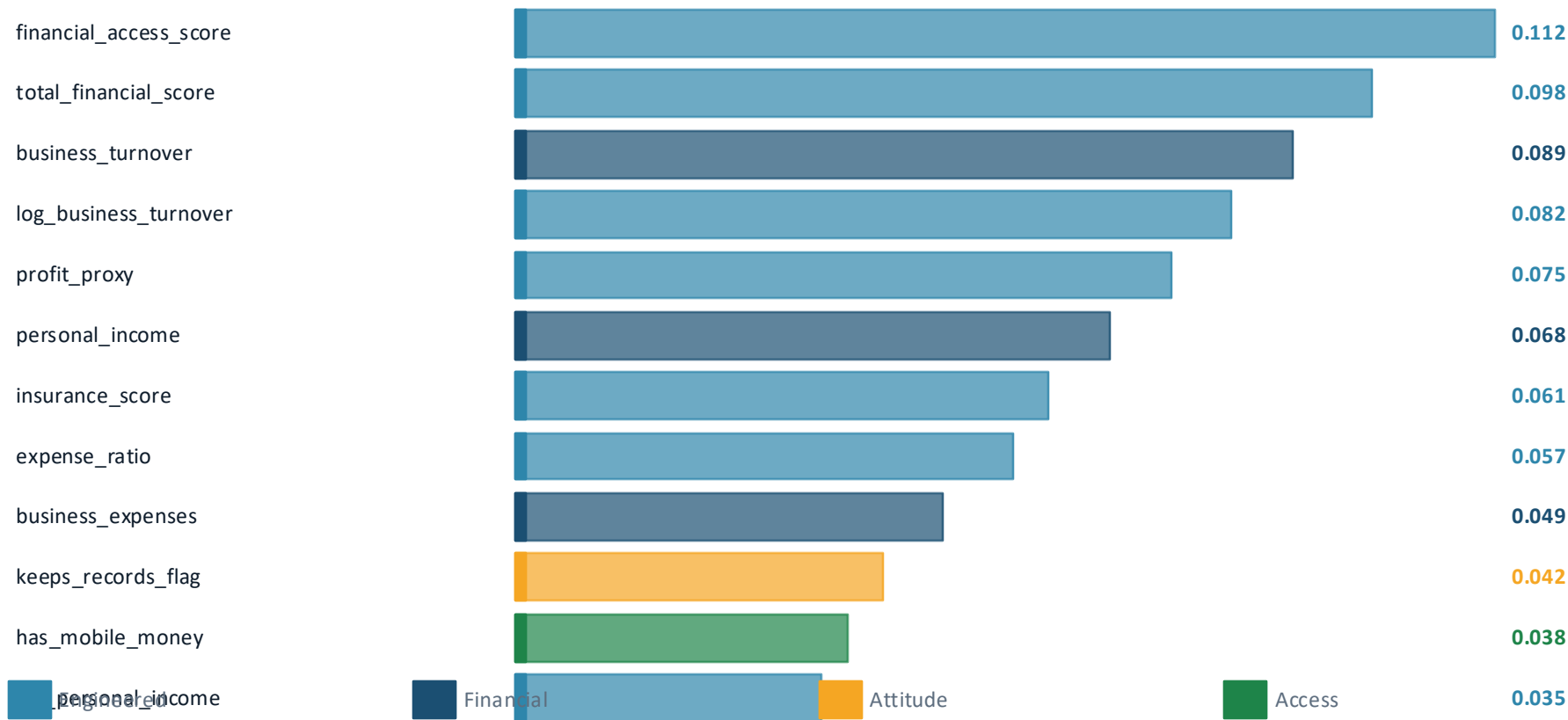


Per-Class F1 (Final Model):



Top Feature Importances

What drives SME financial health predictions? (Random Forest importance scores)



Real-World Business Impact

Who benefits — and how does this model change the landscape?

Financial Institutions



Replace subjective credit assessments



Quantify SME risk with 89% accuracy



Expand loan portfolios safely



Reduce non-performing loan rates

Governments & NGOs



Identify most vulnerable SMEs



Target subsidies and grants effectively



Monitor financial inclusion progress



Design evidence-based policies

SMEs Themselves



Access credit they previously couldn't



Receive tailored financial education



Connect with appropriate support programs



Build financial resilience over time

Key Learnings & Challenges

What we discovered — and what made the difference

Domain features > raw data

Creating financial_access_score, insurance_score, and profit_proxy contributed more than any raw column. Domain knowledge is irreplaceable.

SMOTE placement matters

Applying SMOTE inside folds (not before CV split) was critical to avoiding data leakage and getting honest validation scores.

Class imbalance was the key challenge

'High' health businesses were only ~12% of data. Without SMOTE + class_weight, the model ignored them completely.

Gradient Boosting leads

GB (40% weight) consistently outperformed RF and ET on minority classes — its sequential error correction helps the hardest samples.

Country-grouped imputation

Filling missing values using country-level medians/modes preserved regional economic context better than global imputation.

Ensemble diversity pays off

Three different algorithms each contributed — the ensemble beat every single model on cross-validation by ~1-2% F1.

Conclusion

By combining domain expertise, careful feature engineering, and a diverse ensemble — we achieved an 89% F1 score predicting SME financial health across 4 Southern African nations.

- Financial health goes beyond profit — resilience, savings, and access matter equally

- Machine learning can democratize credit access for millions of underserved businesses

- Domain-engineered features outperformed raw survey variables in all tests

- SMOTE + stratified cross-validation provided honest, leakage-free evaluation

- This approach is replicable across any developing economy with survey data